Positive shared-support distinct-other composition: independent review

The source-level composition matches the retained branch order. It first assumes rawK=0, then that the cancellation equality is false, and finally that the two scaled support coordinates differ. The main source_branch_preserves theorem derives the positive ordinate equality from the two original normalized validity premises; that equality is not an extra hypothesis of the main theorem.

With equality and cancellation false, the common ordinate cannot equal its negative. The proof derives 2≠0 and a.y≠0. This correctly excludes characteristic two and zero-ordinate cases from the positive-only branch; it does not merely add characteristic restrictions to the theorem. The earlier cancellation branch retains those cases.

The distinct source guard a.x*b.z^2≠other.x is converted to distinct normalized x coordinates using nonzero weights. The doubled original has normalized support (X-x_a)^2. Evaluating that support at x_other yields (x_other-x_a)^2≠0. Existing weighted scaling then proves that genericK(other,double(a)) is nonzero. This matches the final source call, including operand order. It excludes the tripling branch without asserting anything about tripling output.

The output-validity proof obtains validity of the other point from shared-support factorization, derives the original point equation from its divisor-validity premise, applies the reviewed doubling theorem, and applies generic addition to the other point and doubled original. Output nonzero weights and normalized Mumford divisibility are conclusions. No output identity or output-validity assumption is supplied. The auxiliary positive_distinct_preserves helper accepts positive equality, but the top-level source theorem discharges it from the original premises.

Source scope is the retained field expressions: generic addition232–292, doubling477–510, mixed dispatcher828–878. The curve is monic degree five with no X4 term. Four representation weights must be nonzero; both original inputs must satisfy normalized validity; rawK must be zero; cancellation must be false; the other support must be distinct. No global nonsingularity premise is needed for this polynomial claim. There is no divisor-class-sum, UInt64, compiler, operation-counter, tripling or whole-walk result here.

Fresh private module, producer controls and separate seven-declaration axiom readback passed. Readback uses only propext, Classical.choice and Quot.sound (positive_nonzero needs no Classical.choice). Producer nonunit rational control invokes the complete source theorem and separately checks normalized output(3/4,5/4,7/8,-3/8). Producer boundary controls retain zero-ordinate cancellation, coincident-support passage and an invalid-point negative control. Independent finite-field controls and their terminal receipts are recorded separately in review.json.

All producer manifest entries and all seven declared source/dependency bindings were checked before replay. Existing pinned imported dependency binaries were reused, not independently kernel-rechecked as binaries or rebuilt transitively. The reviewer previously authored the Doubling dependency, which has its own independent review; the reviewer did not author this positive-branch composition. The producer packet remained untouched. Reviewer preparation and control failures are retained and distinguished from producer discrepancies.

Judgment: supported within the stated field-value and polynomial-validity scope, with no blocking discrepancy. This review provides evidence to the owning coordinator and makes no official research-status or performance decision.